Azure Load Balancer Health Check Troubleshooting

VM-Series Firewall — Internal and External Load Balancer Health Probe Resolution

This guide walks through every layer that must be correctly configured for Azure Load Balancer health probes to reach VM-Series firewalls and receive a healthy response. Azure uses the special IP address `168.63.129.16` for all health probes — if any layer blocks or fails to respond to traffic from this IP, the backend pool member will be marked **unhealthy** and stop receiving traffic.

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How Azure Health Probes Work

Azure Load Balancer sends health probes from the virtual IP `168.63.129.16` to each backend pool member. This is not a routable internet IP — it is an Azure platform IP that exists inside every VNet. The probe arrives on the NIC associated with the backend pool (trust NIC for internal LB, untrust NIC for external LB).

```
Azure Fabric (168.63.129.16)
↓ health probe (TCP/HTTP/HTTPS)
NSG on subnet / NIC
↓
VM-Series NIC (trust or untrust)
↓
PAN-OS interface → zone → security policy → management profile
↓
Response back to 168.63.129.16
```

> **Both LBs probe independently**
>
> The **internal (trust) LB** probes the trust NIC and the **external (untrust) LB** probes the untrust NIC. Each requires its own management profile, security policy, routes, and NSG rules. Fixing one does not fix the other.

Quick Diagnostic Checklist

Run through this list first — most failures are in the top 5

- [ ] **Management profile** on trust/untrust interface allows the probe protocol (HTTP or HTTPS)?
- [ ] **Security policy** allows `168.63.129.16` to the trust/untrust **subnet** on the probe port?
- [ ] **Virtual router has a route** to `168.63.129.16/32` pointing to the correct interface gateway?
- [ ] **NSG** on the trust/untrust subnet allows inbound from `168.63.129.16`?
- [ ] **HA Ports** enabled on the LB rule (required for NVA deployments)?
- [ ] **Floating IP** (Direct Server Return) enabled on the LB rule?

- [ ] **IP forwarding** enabled on the NIC in Azure?
- [ ] **Gateway IP** template variables are set correctly for each AZ/stack?
- [ ] **Backend pool** members reference the correct NIC (trust for internal, untrust for external)?
- [ ] **Firewall has finished bootstrapping** (wait 10–15 min after VM creation)?

Section 1: Firewall Configuration

1.1 Interface Management Profile

The health probe hits the firewall's data-plane interface (not the management interface). PAN-OS will only respond if the interface has a **management profile** that permits traffic from `168.63.129.16` on the probe protocol and port. The management profile must match the LB probe protocol.

LB Type	Interface	Required Management Profile Setting
Internal LB (trust)	ethernet1/2 (trust)	Enable **HTTP** (port 80) or **HTTPS** (port 443) — must match LB probe protocol and permit traffic from `168.63.129.16`
External LB (untrust)	ethernet1/1 (untrust)	Enable **HTTP** (port 80) or **HTTPS** (port 443) — must match LB probe protocol and permit traffic from `168.63.129.16`

How to verify

```
# Check which management profile is attached to the interface
show interface ethernet1/1  # untrust
show interface ethernet1/2  # trust

# View management profile settings
show network interface-management-profile
```

How to configure (Panorama)

1. Navigate to **Network > Network Profiles > Interface Mgmt**
2. Create or edit a profile (e.g., `AZURE-PROBE-HTTP`)
3. Enable **HTTP** (or HTTPS if your probe uses 443)
4. Attach the profile to the correct interface under **Network > Interfaces > ethernet1/x > Advanced > Management Profile**
5. Commit and push

> **HTTPS probes and the health check path**
>
> If using HTTPS probes, the probe path must be `/unauth/php/health.php` (the PAN-OS unauthenticated health endpoint). An HTTP probe on port 80 does not require a path — PAN-OS responds to any HTTP request on port 80 if the management profile allows it. Most deployments use **HTTP on port 80** for simplicity.

1.2 Security Policy

Even with the management profile set, PAN-OS will drop the probe if no security policy permits it. You need an explicit **allow** rule for the Azure health probe IP.

Field	Internal LB Rule	External LB Rule
Name	`AZURE-PROBE-TRUST`	`AZURE-PROBE-UNTRUST`
Source Zone	TRUST	UNTRUST
Source Address	`168.63.129.16` (use address object `H-168.63.129.16`)	`168.63.129.16`
Destination Zone	TRUST	UNTRUST
Destination Address	Trust **subnet** (e.g., `10.51.1.0/24`) — firewalls may be assigned any IP in the subnet	Untrust **subnet** (e.g., `10.51.0.0/24`) — firewalls may be assigned any IP in the subnet
Application	`any`	`any`
Service	`service-http` (or custom port)	`service-http`
Action	**Allow**	**Allow**

Intrazone rule

The health probe enters and exits the same zone (trust→trust or untrust→untrust). This is an **intrazone** flow. If your default intrazone action is set to deny, you must have an explicit allow rule. Most deployments have intrazone allow by default, but verify: `show running security-policy` and check the intrazone-default action.

How to verify

```
# Check if the probe traffic matches a policy
test security-policy-match source 168.63.129.16 destination <TRUST-IP> protoco

# Check running policy
show running security-policy | match AZURE-PROBE
```

1.3 Virtual Router Routes to 168.63.129.16

The firewall must know how to route response traffic back to `168.63.129.16`. This IP is reachable via the **subnet gateway** of the interface that receives the probe. If the route is missing, the firewall drops the response.

Single VR deployments

If trust and untrust are in the same virtual router, a single default route (`0.0.0.0/0`) typically covers it. But verify that the route's next hop sends traffic out the correct interface.

Common Model (dual VR) deployments

In the Common Model, trust and untrust are in **separate virtual routers**. Each VR needs its own route to `168.63.129.16`:

Virtual Router	Destination	Next Hop	Interface
UNTRUST-VR (or CM-VR)	`168.63.129.16/32`	Untrust subnet gateway IP	ethernet1/1
TRUST-VR (or CM-VR)	`168.63.129.16/32`	Trust subnet gateway IP	ethernet1/2

> **Most common failure: wrong gateway IP**
>
> The gateway IP is the **first usable address** in the subnet. For example, if the trust subnet is `10.51.1.0/24`, the gateway is `10.51.1.1`. These are often set via **template variables** (e.g., `$TRUST-CM-VR-GW`, `$UNTRUST-CM-VR-GW`) on the template stack. If the

variable is empty or wrong, the route has no valid next hop and the probe response is dropped.

How to verify

```
# Check routing table for 168.63.129.16
show routing route destination 168.63.129.16

# If using dual VRs, check each one
show routing route virtual-router UNTRUST-VR destination 168.63.129.16
show routing route virtual-router TRUST-VR destination 168.63.129.16

# Verify template variable values (on Panorama)
show template-variable name TRUST-CM-VR-GW
show template-variable name UNTRUST-CM-VR-GW
```

How to check template variables (Panorama GUI)

1. Navigate to **Panorama > Templates**
2. Click the **template stack** assigned to the firewall
3. Review the **Variables** tab — verify gateway IPs match the actual Azure subnet gateways
4. Compare against Azure Portal: **Virtual Network > Subnets** — the gateway is always the first IP in the range (x.x.x.1)

1.4 Interface and Zone Configuration

Check	What to look for
Interface is up	`show interface ethernet1/1` and `ethernet1/2` — state should be `up`
Correct IP assigned	Interface IP must match the Azure NIC private IP. If DHCP, verify it pulled the expected address
Correct zone assigned	Trust interface in TRUST zone, untrust in UNTRUST zone
Correct VR assigned	Each interface assigned to the correct virtual router

Section 2: Load Balancer Configuration

2.1 Health Probe Settings

Setting	Recommended Value	Notes
Protocol	**TCP** or **HTTP**	HTTP on port 80 is simplest. If HTTPS, set path to `/unauth/php/health.php`
Port	**80**	Must match the management profile protocol/port on the firewall
Path (HTTP/HTTPS only)	`/` (HTTP) or `/unauth/php/health.php` (HTTPS)	For HTTP probes, any path works. For HTTPS, use the PAN-OS health endpoint
Interval	**5** seconds	How often Azure sends a probe
Unhealthy threshold	**2**	Number of consecutive failures before marking unhealthy

> **Probe port must match management profile**
>
> If the LB probe is set to port 80 (HTTP), the interface management profile must enable **HTTP**. If the probe is on port 443, enable **HTTPS**. A mismatch means the firewall gets the probe but does not respond.

2.2 Load Balancing Rules

For VM-Series (network virtual appliance) deployments, the LB rules have two critical settings that are frequently misconfigured:

Setting	Required Value	Why
HA Ports	**Enabled**	Forwards ALL protocols and ports to the firewall, not just specific ones. Without this, only traffic matching the rule's port/protocol reaches the firewall. Required for NVA deployments.
Floating IP (Direct Server Return)	**Enabled**	Preserves the original destination IP through the load balancer. Without this, the destination is rewritten to the backend NIC IP, breaking firewall routing and NAT logic.

HA Ports is the #1 LB misconfiguration

If HA Ports is **disabled**, the LB rule only forwards traffic on the specific port/protocol configured in the rule. Health probes may still work (they use a dedicated probe mechanism), but actual data traffic will not reach the firewall for inspection. Always enable HA Ports for VM-Series deployments.

How to verify in Azure Portal

1. Navigate to **Load Balancer > Load balancing rules**
2. Click each rule and verify:
    - **HA Ports**: Yes
    - **Floating IP**: Enabled
    - **Health probe**: Points to the correct probe (port 80 to the correct NIC)

2.3 Backend Pool Configuration

LB Type	Backend Pool Should Reference	Common Mistake
Internal LB	**Trust NIC** (ethernet1/2) of each VM-Series	Using the management NIC or untrust NIC
External LB	**Untrust NIC** (ethernet1/1) of each VM-Series	Using the management NIC or trust NIC

If the backend pool references the wrong NIC, the probe arrives at the wrong interface and either hits no management profile or the wrong zone's security policy.

2.4 Frontend IP Configuration

LB Type	Frontend IP	Notes
Internal LB	Private IP in the **trust subnet** (e.g., `10.51.1.100`)	This IP is the next hop in spoke UDRs. Must be in the same subnet as the FW trust NICs.
External LB	Public IP address	DNAT targets on the firewall must reference this public IP as the destination.

Section 3: Azure Networking

3.1 Network Security Groups (NSGs)

NSGs must allow the health probe traffic from `168.63.129.16`. Azure has a built-in `AzureLoadBalancer` service tag for this, but if you have custom deny rules, they may override it.

NSG Location	Required Rule
Trust subnet NSG	Allow inbound TCP 80 (or 443) from `168.63.129.16/32` (or service tag `AzureLoadBalancer`)
Untrust subnet NSG	Allow inbound TCP 80 (or 443) from `168.63.129.16/32` (or service tag `AzureLoadBalancer`)

How to verify

1. In Azure Portal, navigate to the trust/untrust **NIC > Effective security rules**
2. Look for a rule that matches source `168.63.129.16`, destination port 80 (or 443), action `Allow`
3. If you see `Deny`, check for custom NSG rules with lower priority numbers that override the default `AllowAzureLoadBalancerInBound` rule

> **Default NSG behavior**
>
> Azure's default NSG rules include `AllowAzureLoadBalancerInBound` at priority 65001, which allows all traffic from the `AzureLoadBalancer` service tag. This rule is only overridden if you create a custom deny rule at a lower priority number. If you are using custom NSGs, verify this rule is not being shadowed.

3.2 IP Forwarding on NICs

Azure drops all forwarded traffic (traffic where the destination IP doesn't match the NIC's IP) unless **IP forwarding** is enabled on the NIC. This must be enabled on **both** the trust and untrust NICs.

How to verify

1. Azure Portal > VM > **Networking** > select the NIC
2. Check **IP forwarding**: must be `Enabled`
3. Repeat for both trust and untrust NICs

> **IP forwarding vs health probes**

> IP forwarding does not directly affect health probes (probes target the NIC's own IP). However, if IP forwarding is disabled, data traffic won't flow even when probes pass. Always verify it's enabled as part of the troubleshooting process.

3.3 Terraform Variables to Double-Check

If the deployment was done via Terraform (SWFW modules), these are the variables that directly impact health probe success:

Variable / Setting	Where	What to Check
`enable_ip_forwarding = true`	VM-Series NIC config	Must be `true` on trust and untrust NICs
`health_check_port`	LB module config	Must match the management profile protocol (usually 80)
`ha_ports_rule`	LB rule config	Must be `true` for NVA deployments
`floating_ip`	LB rule config	Must be `true`
NSG ingress rules	Security group config	Must include allow for `168.63.129.16` or `AzureLoadBalancer` tag
Trust/untrust gateway variables	Panorama template stack variables	`$TRUST-CM-VR-GW`, `$UNTRUST-CM-VR-GW` must match Azure subnet gateway (x.x.x.1)

Section 4: Diagnostic CLI Commands

Run these on the VM-Series firewall to diagnose health probe issues:

4.1 Verify probe traffic is arriving

```
# Packet capture on the trust interface for health probe traffic
debug dataplane packet-diag set filter match source 168.63.129.16
debug dataplane packet-diag set filter on
# Wait a few seconds for probes to arrive
debug dataplane packet-diag show setting

# Check session table for probe sessions
show session all filter source 168.63.129.16

# Check traffic log for probe entries
show log traffic source eq 168.63.129.16
```

4.2 Verify interface and routing

```
# Interface status
show interface all

# Check management profiles
show network interface-management-profile

# Routing lookup for probe response
show routing fib virtual-router <VR-NAME> | match 168.63.129.16
test routing fib-lookup virtual-router <VR-NAME> ip 168.63.129.16

# Show all static routes
show routing route type static
```

4.3 Verify security policy match

```
# Test policy match for trust-side probe (use the firewall's actual trust IP)
test security-policy-match source 168.63.129.16 destination <FW-TRUST-IP> \
  protocol 6 destination-port 80 from TRUST to TRUST

# Test policy match for untrust-side probe (use the firewall's actual untrust
test security-policy-match source 168.63.129.16 destination <FW-UNTRUST-IP> \
  protocol 6 destination-port 80 from UNTRUST to UNTRUST

# NOTE: The security policy destination should use the trust/untrust SUBNET
# (not a single IP), since firewalls may be assigned any IP in the subnet.

# Show the running security policy
show running security-policy
```

4.4 Verify counter for drops

```
# Check global counters for drops
show counter global filter severity drop

# Check flow drops specifically
show counter global filter category flow | match drop

# Look for "no-route" or "policy-deny" drops
show counter global filter delta yes | match "no-route\|policy-deny\|icmp-unre
```

Troubleshooting Decision Tree

Step 1: Is the firewall fully booted?

Check `show system info` — if uptime is less than 10 minutes and the firewall just deployed, wait for bootstrap to complete. During bootstrap, the dataplane is not ready and cannot respond to probes.

Step 2: Are probes reaching the firewall?

Check `show session all filter source 168.63.129.16`. If **no sessions**, the probe is blocked before reaching PAN-OS — check NSGs and backend pool NIC assignment. If sessions exist but show `DISCARD`, check security policy.

Step 3: Is the management profile correct?

If sessions show but the LB still reports unhealthy, the firewall is receiving the probe but not responding. Verify the management profile on the probed interface enables the correct protocol (HTTP/HTTPS) and port (80/443).

Step 4: Can the firewall route the response?

Run `test routing fib-lookup virtual-router <VR> ip 168.63.129.16`. If this returns `no route` or routes via the wrong interface, add a static route for `168.63.129.16/32` via the correct subnet gateway.

Step 5: Are the LB rule settings correct?

Even if health probes pass, traffic won't flow without HA Ports and Floating IP. Verify both are enabled on every LB rule.

Palo Alto Networks — Implementation Guides — Azure VM-Series LB Health Check Troubleshooting

For the full deployment guide, see: [VM-Series Deployment on Azure]